Alexander Nguyen

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EDUCATION

Purdue University August 2025 - May 2029

West Lafayette, IN - B.S. in Electrical Engineering GPA: 4.00/4.00

EXPERIENCE

IEEE MTTS (Microwave Theory and Technology Society) September 2026 - Present

RF Front-End Hardware Team Member - Purdue University

- Created end-to-end PCB designs for synthetic-aperture-radar (SAR), including PLLs and PAs in Altium Designer
- Optimized PLL phase-noise performance for a 5.6 GHz radar synthesizer by selecting a 200 MHz reference oscillator and analyzing flicker and flat noise contributions to minimize output jitter.
- Implemented impedance-matched transmission lines and length-matched differential pairs to maintain signal integrity across high-frequency circuit connections
- Co-designed low-noise power delivery architectures using proper LDOs, ferrite beads, and local decoupling, evaluating current budgets and voltage rail allocations for multiple sub-teams and components

VIP ARES (Autonomous Racing Engineering Systems) January 2026 - May 2026

NRHDG Drone-Racing Algorithm Team Member - Purdue University

- Engineered a drone software-in-the-loop (SITL) environment integrating ROS 2, ArduPilot, and Gazebo for autonomous path-planning research.
- Developed 3D racing environments with gates and obstacles, integrating simulated flight dynamics and sensor feedback to support repeatable navigation testing.

PROJECTS

Vacuum Tube Output Transformer-Less Headphone Amplifier / Pre-Amplifier April 2026 - June 2026

Two-stage valve amplifier using 6SN7 and ECC99 tubes

- Engineered a custom power supply including high-voltage DC B+ rails and high-current AC heater supplies
- Ran LTspice transient simulations to optimize biasing, signal linearity, and component values
- Implemented variable global and local negative feedback loops to decrease harmonic distortion from 2.6% to 0.7% and allow fine-tuned voltage gain
- Designed the amplifier's cathode-follower output stage to optimize for headphone loads, measuring a minimum output impedance of 22 ohms

Analog Audio Synthesizer July 2026 - Present

Modular analog synthesizer including voltage-controlled oscillators, filters, and amplifiers

- Designed a modular analog synthesizer integrating voltage-controlled oscillators, filters, and amplifiers to generate and shape audio signals across a targeted 20 Hz–20 kHz range.
- Developed LM13700 OTA-based voltage-controlled filters for adjustable frequency response and resonance, alongside AS3360 amplifiers for dynamic volume control.
- Prototyped and characterized analog circuits using oscilloscopes, measuring voltage gain, cutoff frequencies, and output distortion against design specifications.

EMPLOYMENT

Server May 2026 - Present

Simply Thai - Louisville, KY

- Managed a 6-table section in a high-volume Thai restaurant, handling fast-paced dinner rushes and large group tables

SKILLS

Software: Altium Designer, KiCad, LTspice, Fusion 360, ROS 2, Gazebo, Git, Linux (Ubuntu)
Hardware: PCB Design & Assembly, Analog/RF Circuit Design, Circuit Simulation, Soldering, Oscilloscope, Multimeter
Programming: C (Embedded), MATLAB, ArduPilot SITL, Microsoft Excel, LaTeX